# Metrics in field extraction

Some quick notes

# Motivating example

1. If you use this definition from here:

- **True positive (TP)**: The field exists in the ground truth and our system correctly extracted its value according to the field-specific comparator
- **False positive (FP)**: Our system extracted a value for a field, but either the field doesn't exist in the ground truth or the extracted value doesn't match the expected value
- **False negative (FN)**: The field exists in the ground truth, but our system failed to extract it
- **True negative (TN)**: The field doesn't exist in the ground truth, and our system correctly didn't extract it

# Motivating example - Invoice number

| Document ID | Ground Truth (Expected) | Prediction (Extracted) | Match/Mismatch |
|---|---|---|---|
| doc1 | **INV-100** | **INV-100** | Match |
| doc2 | **INV-200** | **INV-999** | Mismatch |
| doc3 | **INV-300** | **INV-300** | Match |
| doc4 | **INV-400** | **INV-888** | Mismatch |

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| doc2 | INV-201 | INV-202 | Mismatch |
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**Is this really a high-recall/low-precision model to you?**

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Precision: TP/(TP + FP) = 2/(2+2) = 50%
Recall: TP/(TP + FN) = 2/(2+0) = 100%

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"Our model has low precision, but high recall for invoice number" ⇒ Gives the impression predictions from the model will *usually* contain the right invoice number somewhere

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**BAD**

"Our model has low precision, but [illegible] for invoice number" ⇒ [illegible] the impression predictions from the model will *usually* contain [illegible] number somewhere

# Why is it bad?

1. Optimising recall using those definitions ⇒ Optimising for “detecting” invoice_number but not actually extracting it correctly

# What is a better approach?

1. Insight - a mistake in field extraction should increment *both* FN and FP
    a. Not only did you not detect the correct value, but you wrongly guessed the other value.
2. New definitions:
    a. gt_correct: Field value is in the ground truth, model predicted something, and it is correct
    b. gt_incorrect: Field value is in the ground truth, model predicted something, but it is incorrect
    c. gt_missing: Field value is in the ground truth, model predicted nothing
    d. nogt_incorrect: Field value not in ground truth, model predicted something (so is incorrect)
    e. nogt_correct: Field value not in ground truth, model predicted nothing (so it is correct)
3. New calculations
    a. FP = gt_incorrect + nogt_incorrect
    b. FN = gt_incorrect + gt_missing
    c. TP = gt_correct
    d. TN = nogt_correct

# Other non-obvious lessons

1. Treat these two cases as the same: Model detected a field was there and predicted “None” as the value & model completely missed detecting the field
    a. Why? Because no one cares about the difference - a client/end-user will hardly/never require you to make a distinction
    b. Often a sign of premature optimization - trying to keep these two cases distinct will 100% make your code horrible.